

Fidelity-Utility Paradox in Surrogate-Based RL for HVAC Control

More accurate surrogates expose rougher action surfaces and collapse PPO; hybrid separates smooth rollout from physical censoring.

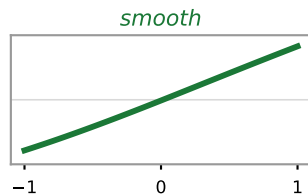
Training backend

Action-response surface

PPO behaviour

Live BOPTEST outcome

Coarse black-box v3
24 h RMSE_T 1.557 °C



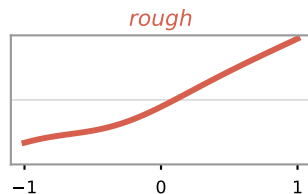
modulating
24% saturation

USABLE

$m_s \approx 0.08$

Accurate single-model
v3.5: RMSE_T 0.644 °C, 7.9x
matched-v3: RMSE_T 0.876 °C, 9.4x

normalised response

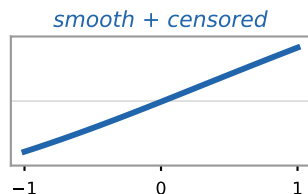


near bang-bang
saturation

COLLAPSE

$m_s > 1$

Hybrid
v3 rollout + frozen v3.5 censor



modulating
~25% saturation

ROBUST

$m_s=0.041$ typ. · viol <5% · 85x faster

RMSE_T = 24 h predictive temperature error · m_s = live maintenance score (lower is better) · violation = fraction of time outside the 21-24 °C comfort band